Hang Le

Washington State University, Ph.D. Candidate, Computer Science

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RESEARCH FOCUS

Ph.D. candidate in Computer Science at Washington State University, with a published paper at **UAI 2026** and an arXiv preprint in **continual learning** and **parameter-efficient fine-tuning** for **LLM agents**. Research spans continual generative memory models, geometry-based skill alignment across agents, and scalable adapter retrieval. Prior industry experience with 3+ years includes production ML systems and large-scale data pipelines at Momo, VNG, and Zalo.

EDUCATION

Washington State University **Expected Graduation:**
Ph.D., Computer Science, GPA: 3.95 / 4.00. **12/2028**

Relevant Coursework: Data Science, Statistical Computing, Artificial Intelligence, Machine Learning, Computer Vision, and Generative AI.

VNUHCM - University of Science 12/2020 - 06/2023
MSc of Mathematical Foundations of Computer Science, GPA: 9.13 / 10.00 Vietnam

Thesis: English–Vietnamese Cross-Lingual Similarity Estimation using Multilingual Encoders.

VNUHCM - University of Technology 08/2016 - 11/2020
B.E. of Computer Science, GPA: 8.39 /10.00 Vietnam

Thesis: Style Transfer Application for Landscape Images using CycleGAN.

SELECTED PUBLICATIONS

- **Hang Thi-Thuy Le**, N.-Q. Nguyen, L.-H. Nguyen, D. Dinh, M. Hoang, T. N. Hoang. *Task-Free Continual Learning via Order-Invariant Linearized Adaptation and Density-Guided Adapter Routing*. Conference on Uncertainty in Artificial Intelligence (UAI), 2026, Amsterdam.
- **Hang Thi-Thuy Le**, L. M. Bui, M. Hoang, T. N. Hoang. *Continual Fine-Tuning with Provably Accurate and Parameter-Free Task Retrieval*. arXiv preprint, 2026.

RESEARCH EXPERIENCE

Research Assistant 01/2024 – Present
WSU, *Machine Learning Lab* (Prof. Trong Nghia Hoang) Pullman, WA

- Proposed order-invariant **linearized PEFT** for task-free continual learning, using closed-form ridge updates over accumulated sufficient statistics and density-guided adapter routing to accommodate non-stationary data distribution shifts; reduced forgetting by **5–7×** versus strong baselines while training fewer parameters (**HESTIA, UAI 2026**).
- Developed parameter-free signature-based retrieval for continual learning, indexing frozen PEFT modules with probabilistic **DP-GMM** signatures and selecting modules through likelihood matching; derived an **exponential retrieval-error bound** and improved retrieval accuracy by up to **57%** over prior retrieval methods across six class-incremental benchmarks (**PROTEUS, arXiv 2026**).
- Designing SIGMA, a signature-indexed parametric memory for LLM agents that consolidates learned adapters into a compact **generative basis** and uses **Gromov-Wasserstein signatures** for sublinear routing and budgeted consolidation, targeting bounded memory growth over open-ended experience; targeting ICML 2027.
- Developing a continual memory model via a probabilistic framework that combines forward **ODEs over statistical geometries** for local skill updates with **Bayesian inference** over a shared latent Poisson point process (PPP) for decentralized alignment; targeting NeurIPS 2027.

INDUSTRY EXPERIENCE

Artificial Intelligence Engineer 07/2021 - 05/2023
Momo Company, *Chatbot Project* Vietnam

- Developed production chatbot intent classification and dialogue models using **Vietnamese transformer architectures**,

improving intent recognition accuracy by **15%**.

- Built a machine learning training pipeline for data collection, annotation, and preprocessing using **Python, BigQuery, and Label Studio**, reducing model iteration time by **40%**.
- Developed evaluation scripts with inter-annotator agreement checks to improve quality control for annotated data.

Data Engineer 07/2020 - 07/2021
VNG Corporation, *Game Data Platform Project* Vietnam

- Built and maintained ETL pipelines for game analytics (**500GB/day**) using **Airflow, Spark, and Python**, ensuring **97% data availability** for downstream analytics.
- Integrated third-party ad data (Facebook, Google, TikTok) and visualized key metrics using Metabase and Apache Hive for data-driven marketing decisions.
- Processed real-time user game data with Kafka and Spark Streaming, cutting latency from hours to minutes.

Data Scientist Intern 04/2019 - 03/2020
Zalo, *Friend Recommendation Project* Vietnam

- Explored friend recommendation prototypes on sampled datasets (from real data with over **70 million users**); engineered features and compared candidate ranking approaches.
- Built Spark/Spark SQL workflows for data extraction, joins/aggregations, and feature engineering.
- Evaluated prototypes with offline metrics (e.g., precision@k/recall@k) and documented findings for team iteration.

ADDITIONAL PUBLICATIONS

- X. Q. Nguyen, Q. T. Nguyen, **H. T.-T. Le**, and D. Dinh. Nôm text recognition using an end-to-end transformer-based architecture with pre-trained models. FAIR Conference, 2023.
- S. Cao, H. Vo, **H. T.-T. Le**, and D. Dinh. Hybrid approach for text similarity detection in Vietnamese based on sentence-bert and wordnet. ITCC Conference, 2022.
- **L. T. T. Hang** and D. Dien. English-vietnamese cross-lingual similarity estimation using language-agnostic representation from multilingual sentence encoders. ICCCI Conference, 2022.
- K. H. Nguyen, D. C. Dinh, **H. T.-T. Le**, D. Dinh. English-vietnamese cross-lingual semantic textual similarity using sentence transformer model. KSE Conference, 2022.

TECHNICAL SKILLS

Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace Transformers, LangChain.

Machine Learning & AI: Continual/Lifelong Learning, Parameter-Efficient Fine-Tuning, LoRA, IA3, Transformers, LLMs, LLM Agents, Deep Learning, Probabilistic Modeling, GMM/DP-GMM, Multimodal Learning, Predictive Modeling.

LLM & AI Models: Llama, Qwen, DeepSeek, T5, CLIP, ViT, GANs.

Programming: Python, SQL (Spark SQL/BigQuery/Hive), R, C++, Bash; Git.

Data & Distributed Systems: Apache Spark, Spark Streaming, Kafka, Airflow, BigQuery, Hive, ETL.

ML/Data Tools: Experiment Tracking, Evaluation Pipelines, Metabase, Label Studio.

AWARDS & ACADEMIC SERVICE

AWARDS

- Outstanding Teaching Assistant, Washington State University 2026.
- Four Scholarships for Academic Excellence, Washington State University, 2024–2025.
- Best paper award, FAIR Conference, 2023.
- Top Students in Computer Science, VNUHCM - University of Technology, 2018–2019.
- Microsoft YouthSpark Scholarship for Outstanding Women Student in Vietnam, 2017-2019.

ACADEMIC SERVICE

- Area Chair: UAI 2026.
- Conference Reviewer: NeurIPS (2026), UAI (2024, 2025).
- Workshop Reviewer: ICML, ICLR, CVPR Workshops (2026)